PhD Research Proposal

Mobile Language Learning Design Environment (MLLDE): a design-based framework for gamified short-form mobile language lessons — prototyping and evaluating *Language Passport*

Applicant: Ana Eslava-Graterol (ProfeAnaEslavaEdTech) **Doctoral programme:** *Languages, Literatures and Cultures and Their Applications* — Universitat Politècnica de València (Department of Applied Linguistics) **Research group:** *Grupo de Investigación para la Enseñanza de Lenguas Asistida por Ordenador* (Computer-Assisted Language Teaching research group), UPV **Research lines:** *Aprendizaje de lenguas asistido por ordenador* (Computer-Assisted Language Learning, CALL) · *E-language learning* **Proposed supervisor (to confirm):** a member of the CALL research group — e.g. Prof. Mª Luisa Carrió-Pastor (Dept. of Applied Linguistics, UPV) *[to confirm]* **Modality / duration:** *[full-time 3 yr / part-time 5 yr — to confirm]* · **Date:** 2026

> *Note.* This is a research-proposal draft prepared by the applicant. It should be read against the programme's official admission requirements and research lines, which the applicant will confirm with the coordinator.

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1. Abstract

Mainstream mobile language-learning applications achieve high engagement but are frequently criticised for privileging decontextualised vocabulary recognition over **communicative and pragmatic competence**, and for lacking an explicit, research-grounded design rationale that teachers can adapt. This project proposes and empirically evaluates the **Mobile Language Learning Design Environment (MLLDE)**: an original instructional- and curriculum-design framework that synthesises established theories — mobile learning design (Clothier), Gamified CLIL and the Input–Processing– Output cycle (Casañ-Pitarch), the CLIL 4Cs (Coyle), Engage–Study–Activate (Harmer), Contextual Teaching

and Learning / REACT (Johnson; CORD), multimedia/dual-coding (Mayer), the Lexical Approach (Lewis), and interlanguage pragmatics (Kasper & Rose) — into a coherent method for **short-form mobile lessons (typically 5–15 minutes)** that satisfy mobile UX constraints and integrate responsible human–AI tutoring. The framework is instantiated in **Language Passport**, a gamified "travel-quest" prototype (already built as a proof of concept). Using **Design-Based Research** operationalised through the **Successive Approximation Model**, the study will iteratively refine MLLDE and measure its effects on learners' communicative/pragmatic performance, self-efficacy, engagement, and active recall, and will assess the framework's **replicability** across destinations and target languages. The intended contributions are a validated design framework, a set of empirically grounded design principles, an open authoring method that lets educators create lessons without programming, and evidence for the application of languages research to educational technology.

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2. Introduction and rationale

Smartphones are now the primary gateway to informal language study, yet the dominant apps optimise for streaks and retention metrics more than for the ability to *act* in the language in authentic situations. Teachers and applied linguists, meanwhile, hold robust pedagogical theory but lack a fast, principled way to translate it into engaging mobile experiences. This gap — between **sound second-language pedagogy** and **effective mobile UX** — motivates the present research.

The applicant's prior work (an MA thesis on disciplinary variation in metadiscourse, UPV, Department of Applied Linguistics) and professional practice in instructional design converge on a single question: *how should mobile language lessons be designed so that they are simultaneously pedagogically rigorous, motivating, and buildable by educators?* MLLDE is the proposed answer, and *Language Passport* is its first working instantiation.

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3. State of the art

- **Mobile-assisted language learning (MALL).** MALL research documents affordances of ubiquity, personalisation, and micro-sessions, but also fragmentation and shallow practice (Kukulska-Hulme & Shield, 2008; Stockwell, 2010; Godwin-Jones, 2011).

- **Microlearning & mobile learning design.** Clothier (2026) argues mobile learning needs its *own* design logic — context-driven, cognitive-load-managed, Bloom-based objectives — rather than shrunken e-learning.
- **Gamification & serious games.** Gamification can increase motivation when mechanics are tied to learning goals (Deterding, Dixon, Khaled & Nacke, 2011); Casañ-Pitarch's **G-CLIL** four-ring model binds game mechanics to disciplinary language tasks.
- **CLIL & communicative/task-based pedagogy.** Coyle, Hood & Marsh (2010) (4Cs); Harmer (2007) (ESA) provide content-and-communication and sequencing logics.
- **Contextual & authentic assessment.** Johnson (2002) and the REACT strategies (CORD; Crawford, 2001) foreground meaningful, performance-based learning.
- **Cognition & lexis.** Mayer (2009) (multimedia/dual-coding); Lewis (1993) (lexical chunks) inform vocabulary and hinting design.
- **Pragmatics.** Interlanguage-pragmatics research (Kasper & Rose, 2001) shows pragmatic competence is teachable yet under-served by apps.
- **Generative AI tutoring.** Emerging work explores LLM tutors; a key open problem is **scaffolding without giving answers** (responsible human–AI collaboration).

Synthesis gap. These strands are rarely integrated into one design method built *for* the short-form mobile constraint, nor evaluated as a replicable, teacher- authorable environment. MLLDE addresses this gap.

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4. Research questions and hypotheses

- **RQ1 (design).** What design principles constitute an effective MLLDE for 5–15-minute mobile language lessons that integrate mobile UX, CLIL/IPO pedagogy, gamification, and AI-mentor scaffolding?
- **RQ2 (outcomes).** How does a MLLDE-designed lesson (*Language Passport*) affect learners' **communicative and pragmatic task performance** and **self-efficacy** compared with a baseline condition?
- **RQ3 (mechanisms).** How do (a) gamified authentic assessment, (b) AI-mentor graded hints, and (c) dual-coded **animated hints** influence engagement and active recall?
- **RQ4 (transferability).** To what extent is the MLLDE method **replicable** across destinations and target languages via an *engine + data* authoring model?

Hypotheses (illustrative): H1 — MLLDE lessons yield higher gains in an authentic speaking/ordering task than a vocabulary-drill baseline; H2 — the animated-hint (same-

image) condition improves recall and reduces cognitive load versus a text-hint condition; H3 — AI-mentor scaffolding that withholds answers increases learner production over an answer-giving condition.

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5. Aims and objectives

General aim. To develop, formalise, and empirically validate the MLLDE framework and its design principles for gamified micro-lessons in mobile language learning.

Specific objectives. 1. Formalise MLLDE (constructs, layers, the Destination Schema) from the theory synthesis. *(RQ1)* 2. Build and iteratively refine *Language Passport* as the design artifact. *(RQ1)* 3. Measure learning outcomes, self-efficacy, engagement, and usability. *(RQ2–RQ3)* 4. Isolate the effect of key mechanisms (animated hints; AI scaffolding; authentic assessment) through focused comparisons. *(RQ3)* 5. Test replicability by authoring additional destinations/languages and evaluating design effort and fidelity. *(RQ4)* 6. Produce validated design guidelines and an authoring method for educators.

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6. Theoretical framework

MLLDE nests established frameworks rather than treating them as alternatives:

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SAM  ── how the artifact is BUILT (agile; Preparation · Savvy Start · Alpha/Beta/Gold)
 └─ G-CLIL Four Rings ── how the lesson is STRUCTURED (mechanics = language tasks)
    └─ IPO × ESA ── how each micro-session FLOWS (input → processing → output)
       └─ CTL + REACT ── how learning stays MEANINGFUL & AUTHENTIC
          └─ Mobile UX (Clothier) + multimedia/dual-coding (Mayer) + pragmatics
             + responsible human–AI collaboration ── how it FEELS and TUTORS on a phone
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The synthesis is the study’s theoretical contribution; the individual frameworks remain attributed to their authors.

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7. Methodology

Design. Design-Based Research (DBR) (Reeves; Anderson & Shattuck, 2012): iterative development and study of a design intervention in authentic contexts to produce both a working artifact and transferable design theory. Build cycles follow the **Successive Approximation Model** (Allen & Sites, 2012): Preparation → Iterative Design (Savvy Start) → Iterative Development (Alpha/Beta/Gold).

Phases. 1. **Analysis & design** — formalise MLLDE; expert validation of the framework and the Destination Schema (Delphi / expert panel of applied linguists and designers). 2. **Development** — refine *Language Passport* (already prototyped: Barcelona, Sevilla; Grace AI mentor). 3. **Iterative evaluation** — pilot → alpha → beta cycles with learners. 4. **Replication study** — author new destinations/languages; measure design effort and pedagogical fidelity.

Participants. Adult A1–A2 learners of Spanish (and, in the replication phase, of a second target language), recruited via UPV language services / partner language schools; plus an expert panel. Sample sizes and sampling to be finalised with the supervisor and ethics board.

Design comparisons (RQ3). Within-/between-subjects micro-experiments contrasting: animated-image hints vs. text hints; AI scaffolding vs. answer-giving; authentic-mission assessment vs. multiple-choice.

Instruments & data. - **Performance:** CEFR-aligned **speaking/ordering task** rubric; a written-production task; pragmatic-appropriateness ratings. - **Affective:** language-learning **self-efficacy** and motivation scales. - **Engagement/behaviour:** app telemetry (completion, coins, retries, time-on-task). - **Usability:** System Usability Scale (SUS); cognitive-load measures. - **Qualitative:** semi-structured interviews; think-aloud; expert reviews.

Analysis. Mixed methods — inferential statistics (pre/post, between-condition effects) triangulated with thematic analysis of qualitative data.

Rigour. Iterative member-checking, expert validation, inter-rater reliability on rubrics, and DBR’s emphasis on ecological validity.

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8. Work plan (indicative)

Year	Focus	Milestones
1	Literature, formalise MLLDE, expert validation; refine prototype	Framework paper; validated Destination Schema; ethics approval
2	Iterative learner studies (pilot → alpha → beta); mechanism experiments	Outcome + mechanism papers; refined design principles
3	Replication (new destinations/languages); synthesis; writing	Replicability paper; guidelines; thesis; defence

(A part-time schedule extends this across ~5 years.)

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9. Expected results and contributions

- **Theoretical:** the **MLLDE** framework and a set of empirically grounded design principles for mobile micro-lessons — a contribution to MALL, CLIL, gamification, and GenAI-in-education literatures.
- **Practical:** the *Language Passport* application and an **engine + data authoring method** enabling educators to create lessons without programming.
- **Methodological:** a documented DBR × SAM process for language-learning design.
- **For the programme:** a concrete demonstration of *languages and their applications* — applied linguistics informing educational technology and design.

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10. Relevance and fit with the doctoral programme

The project sits squarely within the **Grupo de Investigación para la Enseñanza de Lenguas Asistida por Ordenador** (Computer-Assisted Language Teaching research group) and its two research lines: **Aprendizaje de lenguas asistido por ordenador (CALL)** and **E-language learning**. Mobile-assisted language learning (MALL) — the terrain of *Language Passport* — is a core, contemporary sub-field of CALL, and the study's gamified, AI-mediated, mobile micro-lessons are a direct instance of e-language learning.

More broadly, the programme's focus on **languages, literatures and cultures *and their applications*** encompasses applied-linguistics research that produces usable artefacts and methods. MLLDE contributes to that *applications* strand: it applies second-language-acquisition and CLIL theory, discourse/pragmatics, and instructional design to a real technological product — aligning with the CALL group's expertise, the Department of Applied Linguistics, and the applicant's prior UPV training.

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11. Ethical considerations

Informed consent; voluntary participation; data minimisation and GDPR compliance (pseudonymised telemetry; client-side storage where possible); safeguarding in any AI-mediated interaction (the AI mentor is constrained to educational, age-appropriate tutoring). Ethics approval will be sought from the UPV committee before data collection. Third-party pedagogical frameworks are credited to their authors; the applicant's original contributions are protected (proprietary licence; "Language Passport"™).

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12. Preliminary results (proof of concept)

A working prototype already evidences feasibility: *Language Passport* runs the full MLLDE lesson cycle (arrival → animated vocabulary → lexical chunks → model dialogue → gap-fill → authentic NPC mission → oral/written reflection → culture & pragmatics cards → rewards), with a persistent gamified economy, a world-tour map, offline PWA delivery, and an AI game master (**Grace**, deployed as a custom GPT). Two destinations (Barcelona, Sevilla) are playable, demonstrating the *engine + data* replicability the study will test. *(A "before/after" redesign and the full design record are documented in the project repository.)*

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13. Resources

Standard doctoral resources: UPV library/databases; access to learner cohorts via UPV language services / partner schools; the applicant's existing prototype and design toolkit;

standard statistical and qualitative-analysis software. No specialised equipment beyond mobile devices is required.

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14. Selected references (indicative — to be expanded/verified)

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(Bibliographic details to be completed and formatted to the programme's required style.)

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